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Stored Procedure

A stored procedure is a collection of SQL statements saved inside a database that we can execute whenever we need them. Think of it like a saved recipe. Instead of writing the same SQL query again & again, we save it once and call it when ever we want

for ex, if we frequently need to display all employes, we wan create a stored procedure for that task.

Why do we use Stored procedure?

- Reusability: write SQL code once and reuse it
- Performance: can reduce repeated processing and improve efficiency in suitable cases
- Security: Access can be granted to execute a procedure without granting direct access to all underlaying tables
- Reduce network traffic: Send a procedure call instead of sending a long SQL query each time
- Easy maintenance: update the logic in one place

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Syntax : DELIMITER //
CREATE PROCEDURE procedure_name()
BEGIN
SELECT * FROM emp;
END //
DELIMITER ;

DELIMITER //: temporarily changes the statement ending symbol to //
CREATE PROCEDURE: creates a stored procedure
procedure_name(): Name of the procedure; here no parameters are used
BEGIN : Starts the procedure body
SELECT * FROM emp; : retrieves all columns & rows from the emp table
END // : Ends the procedure definition
DELIMITER ; : Restores the normal semi colon delimiter

DELIMITER: Normally, MySQL uses ; to identify the end of the statement. But a procedure can contain multiple SQL statements, each ending with a semicolon,
so we temporarily change the delimiter to // allowing MySQL to recognize the end of the

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entire procedure definition

Step 1 - Create

Ex: DELIMITER //

```
CREATE PROCEDURE getEmployees ()
BEGIN
    SELECT * FROM emp;
END //
DELIMITER ;
```

Step 2 : Execute

```
CALL getEmployees ();
```

Types of parameters

A parameter allows us to pass information into a procedure or get information back from it

1. IN: Input parameter, we give a value to procedure and it uses that value

Ex: Find the employee whose employee no is 7521

```
DECIMITER //
CREATE PROCEDURE getEmployeeById (IN eid
    INT)
```

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```
BEGIN
 SELECT * FROM emp
 WHERE empno = eid;
END //
DELIMITER;
```

Execute it : CALL getEmployeeById (7521);

How does it works?

1. The value 7521 is passed into the eid parameter
2. The WHERE condition checks empno = eid
3. Only the matching employee is returned

The imp point is that eid is a parameter, while empno is a column in the table

2. OUT : output parameter the procedure calculate a value and returns it

Ex: Return the total no of employees

```
DELIMITER //
CREATE PROCEDURE totalEmployees (OUT
                                 total INT)
BEGIN
  SELECT COUNT(*) INTO total
```

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```
FROM emp;
END//
DELIMITER;
```

Execute it : CALL totalEmployees (@total);
SELECT @@ total;

- COUNT(*): counts the rows in the emp table
- INTO total : stores that count in the procedure's output parameter
- @total : is a MySQL session variable that receives the output
- SELECT @ total ; ; displays the returned value

* total and @total are different. total is the procedure parameter, whereas @total is a session variable

3 INPUT : Input and output parameter, The procedure receives a value, changes it, and returns the updated value

Ex : Receive 3000, add 2000, and return 5000

```
DELIMITER //
CREATE PROCEDURE Salary (INPUT sal DECIMAL
```

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```
(10,2))
BEGIN
  SET Sal = Sal + 2000;
END//
DELIMITER;
```

Execute it : SET @Sal = 3000;
CALL Salary (@Sal);
SELECT @Sal;

How to drop and view stored procedures

a DROP a procedure

```
DROP PROCEDURE getEmployee;
```

Removes the stored procedure named getEmploy from the database. This does not delete the employee table

b SHOW procedure status

```
SHOW PROCEDURE STATUS;
```

Displays information about stored praedures such as their names and databases they belong to

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c SHOW the procedure definition

SHOW CREATE PROCEDURE getEmployees;

Displays the SQL definition used to create getEmployees

NOTE: The procedure must exist before you can drop it or inspect its definition

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